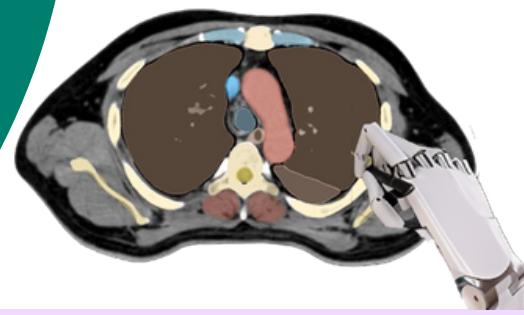


3RD WORKSHOP ON RADIOMICS AND AUTO SEGMENTATION



This two-day workshop is to introduce clinicians, medical physicists and imaging researchers to radiomics and autosegmentation. Participants will learn how to extract and analyze quantitative imaging features to improve diagnosis, prognosis, and treatment planning. The workshop includes hands-on sessions using accessible tools for image segmentation and radiomics feature extraction, model building using machine learning and interpretation, with minimal coding skills required. Participants are requested to bring their own laptops.

KEY TOPICS COVERED

- How to Formulate Questions in Radiomics Research
- Radiomics Fundamentals & Clinical Value
- Radiomics Workflow Overview
- Image Segmentation Techniques
- Radiomics Feature Extraction
- Machine Learning Basics and Predictive Modeling
- Limitations and Critical Interpretation
- Hands-On Practical Sessions on Segmentation, feature extraction, and building predictive models

SAVE THE DATE

18 & 19TH SEPTEMBER 2026

Time : 8:30 Am to 4:30 Pm



Radiation Oncology Unit 2
CMC Vellore, Ranipet

Who Should Attend?

Clinicians (e.g Oncologists, Radiologists, etc) physicists, and medical imaging researchers who are interested in integrating quantitative imaging analytics into health care decision making.

Prerequisites

Basic familiarity with medical imaging (Radiology, DICOM etc) recommended.

Registration

Send an email to cmcbmiuconnect@cmcvellore.ac.in .
First-come first serve basis. **Limited seats available.**
Payment link will be provided and through other means not allowed.
Registration fee: Rs 7500 (Rs 5000 - CMC staff) excluding GST.
Last date of registration **September 9th, 2026**

Laptop Requirements

Minimum: i5 / R5 CPU, 8 GB RAM, 10–20 GB free storage
Recommended: i7 / R7 CPU, 16 GB RAM, 20+ GB free storage, Nvidia GPU
*Laptops can also be provided at rental cost if required

ORGANIZED BY



Christian Medical College
Vellore
Radiation Oncology Unit 2



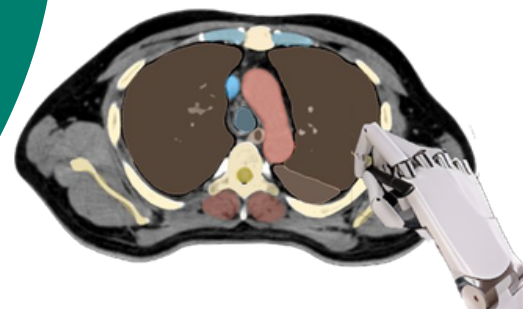
QIRAIL

(Quantitative Imaging
Research and AI Lab)

Biomedical Informatics Unit

3RD WORKSHOP ON RADIOMICS AND AUTO SEGMENTATION

18 & 19 SEPTEMBER 2026



**Venue : 18-Residents room,
B0004 (Basement), Radiation Oncology, Ranipet Campus,
Christian Medical College Vellore**

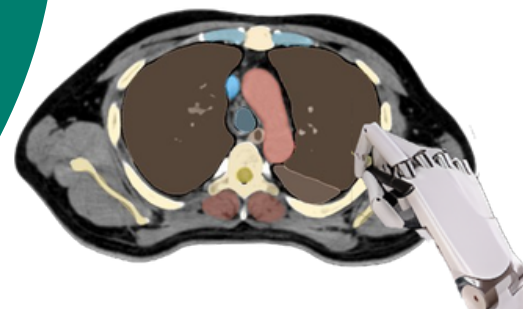
DAY 1 -18/09/2026

Time	Session Title	Facilitator(s)	Key Focus/Description
08:30 AM - 09:00 AM	Registration & Welcome		
09:00 AM - 09:30 AM	Workshop Overview and Introduction to Radiomics	Dr Hannah Mary Thomas	Objectives & Outcomes, Basic Concept & Evolution, DICOM Overview, Types Radiomic Features
09:30 AM - 10:00 AM	Inauguration		Chief guest: Dr Biju George, Director, CMC Vellore
10:00 AM - 10:15 AM	Tea Break		
10:15 AM - 11:00 AM	Research Planning: Formulating a Clinically Meaningful AI Research Question	Dr Balu Krishna S	Identifying a clinically meaningful problem and defining a clear outcome (Hypothesis design)
11:00 AM - 11:45 AM	Getting Started with Radiomics Research: - interactive session	Dr Prasanna Samuel, Biostatistics, Dr Hannah Mary Thomas Moderator: Dr Balu Krishna S	Discuss few participant research questions, Data handling, Sample size considerations in AI studies, Basic statistics used in Machine Learning, Feature selection and IBSI compliance
11:45 AM -12:15 PM	Building Machine-Learning and AI Models for Radiomics	Dr Hannah Mary Thomas	Basics of AI models, Choice of ML classifiers
12:15 PM - 1:00 PM	Lunch Break		
01:00 PM - 2:30 PM	Hands-on 1 : Radiomics feature extraction using IBSI compliant tools	Mr Sathya A	Radiomics extraction using 3DSlicer
02:30 PM - 4:15 PM	Hands-on 2: Model Building	Mr Hasan Shaikh	Data to model building using clinical and radiomics data
04:15 PM - 4:30 PM	Tea Break		

Note : Schedule is subject to change

3RD WORKSHOP ON RADIOMICS AND AUTO SEGMENTATION

18 & 19 SEPTEMBER 2026



**Venue : 18-Residents room,
B0004 (Basement), Radiation Oncology, Ranipet Campus,
Christian Medical College Vellore**

DAY 2 - 19/09/2026

Time	Session Title	Facilitator(s)	Key Focus/Description
08:30 AM - 08:45 AM	Recap	Dr Sathiyaraj P, Kidwai Memorial Institute of Oncology	
08:45 AM - 09:30 AM	Medical Image Segmentation: Methods and Applications	Dr Jeny Rajan, NITK Surathkal	Review of segmentation methods: manual, semi-automatic, and fully automatic
09:30 AM - 10:00 AM	Demonstration of medical Image Segmentation applications	Mr Asjad Nabeel, GEC Wayanad & Mr Sathya	Tools and software platforms commonly used for medical image segmentation
10:00 AM - 10:30 AM	Beyond the Research Question: What makes AI Research Feasible	Dr Balu Krishna S	Learnings from QIRAIL
10:30 AM - 10:45 AM	Tea break		
10:45 AM - 11:15 AM	Evaluating AI Models: Validation and Performance Metrics	Mr Asjad Nabeel, GEC Wayanad	Segmentation and Classification models; Cross validation, Performance metrics
11:15 AM - 11:45 AM	Reporting Guidelines for AI/ML Studies: TRIPOD-AI etc.	Dr Sathiyaraj P Kidwai Memorial Institute of Oncology	Understanding standardized AI/ML reporting frameworks (e.g., TRIPOD-AI, CONSORT-AI, STARD-AI)
11:45 AM - 12:15 PM	Group Activity	Dr Glynis Francis	Critical Appraisal of an AI/ML Research Paper
12:15 PM- 01:00 PM	Lunch and group photo		
01:00 PM - 03:00 PM	Hands-on 3 : Medical Image segmentation	Mr Sathya A	Manual and Automated Image segmentation using 3DSlicer
03:00 PM- 04:00 PM	Hands-on 4 : Model evaluation and interpretation	Mr Hasan Shaikh	Model evaluation and interpretation
04:00 PM - 04:15 PM	Discussion & Vote of Thanks	Dr Hannah Mary Thomas	
04:15 PM- 04:30 PM	Tea break		

Note : Schedule is subject to change